

MuRL-DTI: A Multimodal Feature Fusion Reinforcement Learning Approach for Cold Start in Drug-Target Interactions

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Abstract—Drug Target Interaction (DTI) focuses on exploring the interactions between specific drug molecules and their biological targets to assess the efficacy and safety of drugs. Significant advancements have been made in integrating computational techniques compared to traditional approaches, including machine learning-based DTI models, deep learning-based DTI models, and knowledge graph-based DTI models. However, these models often depend on rich contextual and structural information, which can lead to reduced prediction accuracy when dealing with sparse data and pose challenges in cold start scenarios. To address this limitation, we propose a novel DTI prediction method called MuRL-DTI. Specifically, we employ DQN-based reinforcement learning to the DTI task to solve the cold-start problem by leveraging learned knowledge to transfer to the unknown drug-target interaction. Additionally, we utilize biological information from multiple sources, including sequence features and compound features, to reinforce the learned environmental information and improve the prediction accuracy. Experimental results on three benchmark datasets demonstrate that our model outperforms the baselines, validating the effectiveness of the proposed approach.

Index Terms—Multi-modal, Reinforcement Learning, Cold Start, Drug Target Interaction.

I. INTRODUCTION

Drug-Target Interaction (DTI) prediction [1], [2] involves predicting the interactions between drug molecules and biological targets, such as proteins, using methods from computational biology and bioinformatics. This research aims to accelerate drug development [3], improve success rates, identify new targets [4] and optimize drug design [5]. Ultimately, these efforts support new drug discovery and personalized medicine while reducing the costs and risks of research and development.

Traditional DTI prediction methods are primarily conducted through biochemical experiments, which can be time-consuming and risky. Significant progress has been made in DTI prediction by incorporating computational techniques that efficiently process and analyze large amounts of biological data to improve prediction accuracy. Existing methods can be broadly categorized into three types: machine learning-based [1], deep learning-based [2], and knowledge graph-based models [6].

Machine learning methods for DTI prediction primarily include similarity-based, feature extraction-based, and semi-supervised approaches. The core principle of similarity-based machine learning methods is that similar drugs tend to interact with similar targets and vice versa. Notable models in this category include [7]–[13]. The feature vector-based approach represents drugs and targets as feature

vectors, which are then classified using a classification algorithm. Typical models in this category include [14]–[16]. Semi-supervised methods are primarily employed to mitigate the effects of negative sample selection on DTI predictions, with notable models including [17]–[21].

The core principle of deep learning-based DTI prediction task is to automatically extract and learn high-dimensional feature representations of drugs and targets by constructing a deep neural network model. This approach aims to capture complex interaction patterns and ultimately achieve accurate predictions of potential interactions between drugs and targets. Typical models include GraphDTA [22], BridgeDPI [23], MGraphDTA [24], DeepGLSTM [25], CSatDTA [26], DTITR [27], CoaDTI [28].

Knowledge graphs (KGs) provide a structured representation of drug interactions by capturing the relationships between drugs and their associated properties. For example, TriModel [29] incorporated prior information about both the drug and the target, thereby increasing prediction accuracy. DTiGEMS+ [30] combines drug-drug and target-target similarity maps to improve drug-target interaction maps. The NeoDTI model introduces a novel nonlinear end-to-end learning approach that integrates diverse information from heterogeneous graph data. The EEG-DTI model [31] employed graph convolutional networks (GCN) to predict DTI based on low-dimensional feature representations of drugs and targets within heterogeneous graphs. GraphDTA [22] employed graph neural networks to predict drug-target affinity.

However, existing models depend on rich, contextually informative data for effective modeling and exhibit low prediction accuracy when faced with sparse data and cold-start problems. The cold-start [32] problem refers to the lack of sufficient historical data to reliably predict the interactions of new drugs or targets. For example, the introduction of new drugs or targets can decrease the accuracy of the model due to insufficient contextual data. To address this challenge, we propose a multimodal reinforcement learning (RL) model for DTI prediction, called MuRL-DTI, which leverages learned knowledge about interactions to predict new drug-target combinations. Specifically, we first construct an interaction network by organizing drug and target information from the biological database, connecting interacting drugs and targets by edges. We then employ a DQN-based RL model to learn the current interaction information within the network. This model integrates experience replay and target network

mechanisms to effectively utilize historical knowledge, enabling it to perform on unknown or sparse data. Additionally, we propose the integration of multi-source biosignatures, including sequence features and compound features, into the RL framework to enrich the environmental information and improve the model’s performance. Our contributions are as follows:

- We propose a multimodal fusion reinforcement learning model, called MuRL-DTI, which uses a DQN-based RL model to transfer the acquired knowledge to the interactions of unknown drugs and targets. Additionally, it incorporates multi-source biometric information to enrich the RL environment and improve the prediction accuracy.
- We conduct experiments to evaluate the performance of our model on three publicly available DTI datasets. The results demonstrate that MuRL-DTI outperforms existing models, validating its effectiveness.

II. METHODOLOGY

A. Preliminaries

a) Environments in RL - States, Actions, Transitions, and Rewards: The $\mathcal{S}$ represents the state space, which contains all possible states of the environment. The state $S \in \mathcal{S}$ is represented by $S = (e_t, e_{1q}, r_q, e_{2q})$, where the state space includes all valid combinations. The $\mathcal{O}$ represents the observation space. Since the environment is defined as partially observable, this means that the decision maker does not have full access to all state information of the environment, but only partial information through observations. The observation space defines all possible observations that the decision maker can receive. The $\mathcal{A}$ represents the action space, which contains all possible actions that the decision maker can choose from. By choosing different actions, the decision maker aims at changing the state of the environment in order to achieve a certain goal. δ represents the state transfer function, which describes how the state of the environment changes after a given action. The state transfer function is deterministic, meaning that for a given state and action, the environment will transfer to a specific new state. R represents the reward function, which assigns a numeric reward to each state transfer.

B. Multi-modal Feature Fusion

We fuse two types of information to enhance the environmental context: the protein sequence feature of drugs and the structural feature of compounds. Fig.1 shows the multimodal features that are extracted. The protein sequence can be represented as “MVSY-WDTGGVLLSSS...,” which provides a more accurate reflection of the structure and properties of proteins compared to traditional descriptive texts. To effectively extract features from protein sequences, we utilize a well-established protein feature extraction tool, FECS [33]. This tool performs sequence extraction based on the structural and statistical features of proteins, and its accuracy has been validated in various research tasks.

For the structural feature of compounds, we first employ the SMILES representation [34], which converts the molecular composition of compounds into a string format. We then use the RDKit tool to transform these string representations into structural diagrams. Finally, the structural features of the compounds are generated using the GraphDTA algorithm [22].

C. DQN-Based Multimodal Reinforcement Learning

Fig. 2 shows an overview of DQN-based multimodal reinforcement learning. We first construct the interactions between the drug and the

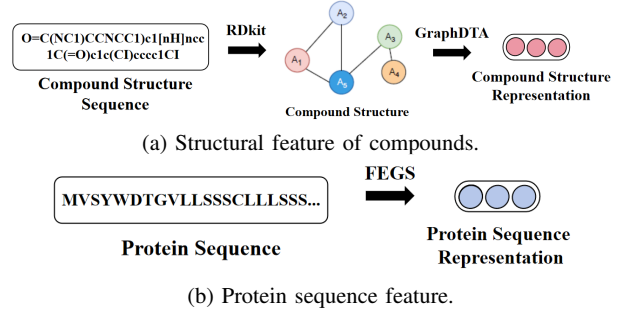


Fig. 1: Multimodal feature extraction.

target in an interaction network and allow the model to learn in this network environment. Then we present a RL model based on Deep Q-Networks (DQN) that takes into account the high-dimensional characteristics of biometric features. Specifically, the model uses two multilayer neural networks with identical architecture, called the $Q_{Prediction}$ and the Q_{Target} network, Fig 3 shows the structure of the network. The primary function of the $Q_{Prediction}$ network is to select optimal actions based on the current state of the environment, while the Q_{Target} network generates the target Q-values for these actions, which represent the expected future rewards. To enhance the stability of the learning process, the DQN approach incorporates a periodic updating mechanism, in which the parameters of the $Q_{Prediction}$ network are periodically transferred to the Q_{Target} network. This procedure mitigates the discrepancies between the target Q-values and the predicted Q-values that arise during training, thereby preventing oscillations and divergence in the learning process.

The model is trained as follows: The input state s_t represents the feature representation of a drug or target. The model outputs Q-values for all possible actions given the state s_t . Using a greedy strategy, an action a_t is selected, which is then input into the environment. This interaction results in a new state s_{t+1} and a reward r_t . The reward r_t is defined based on the outcome of the interaction between the drug and the target, assigning a value of +1 if the drug effectively targets the goal and -1 if it does not. Subsequently, the value of the Q prediction network is computed:

$$y_t = r_t + \gamma \cdot \max_a Q_{Target}(s_{t+1}, a; w) \quad (1)$$

where γ denotes the discount factor that measures future rewards. It controls the weight assigned to future rewards and typically takes a value between 0 and 1. Then calculate the loss function:

$$L = 1/2 [y_t - Q_{Prediction}(s, a; w)]^2 \quad (2)$$

Based on the loss, the parameters of the Q prediction network are updated using gradient descent to minimize the difference between $Q_{Prediction}(s_t, a_t)$ and y_t . Then update Q prediction network as follows:

$$Q_{Prediction}(s_t, a_t, w) \leftarrow Q_{Prediction}(s_t, a_t, w) + \alpha [r_{t+1} + \gamma \max_a \hat{Q}(s_{t+1}, a_t, w) - Q_{Prediction}(s_t, a_t, w)] \quad (3)$$

After completing the above steps, a quaternion transition: (s_t, a_t, r_t, s_{t+1}) is obtained. Repeat the above process until a predetermined number of training repetitions are achieved.

During the update process, only the weights of the prediction network $Q_{Prediction}$ are updated, while the weights of the target

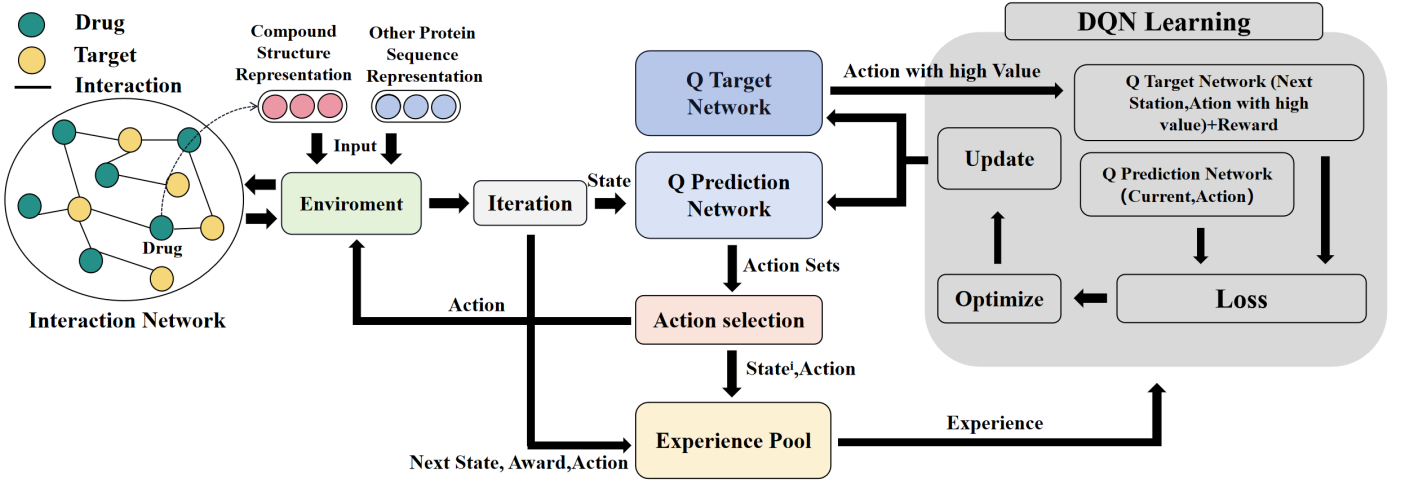


Fig. 2: An overview of DQN-based multimodal reinforcement learning.

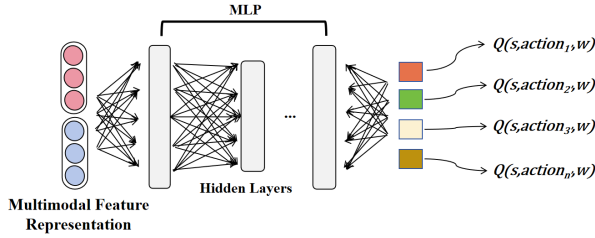


Fig. 3: The structure of the Q prediction and Q target networks.

network Q_{Target} remain unchanged. After a certain number of updates, the weights of the updated $Q_{Prediction}$ are copied to Q_{Target} for the next batch of updates.

III. EXPERIMENTS

A. Experiment Setting

a) *Datasets*: We employ three public datasets for experimental validation, namely Luo's [35], Yamanishi's [36], and Zheng's [37] datasets. Luo's constructs a complex bioinformatic network that includes drug-target, drug-drug, and protein-drug interactions, with the network consisting of 12,015 nodes and 1,895,445 edges. Zheng's contains a wide range of drug and target data, documenting 11,819 drug-target interactions involving 1,094 different drugs and 1,556 targets. Yamanishi's includes known drug-target interactions, the numbers were 635, 2926, 1476, and 90, respectively.

In each dataset, all known DTIs are classified as positive samples, while unrecorded drug-target pairings are classified as negative samples. Additionally, two different ratios are used to select positive samples: 1:1 and 1:10. This means that for each positive sample, an equal and tenfold number of negative samples are selected to create two distinct experimental data sets. This setup is designed to consider the potential impact of sample proportions on the results.

b) *Metrics*: We employ two widely used evaluation metrics: AUC (Area Under the Receiver Operating Characteristic Curve) and AUPR (Area Under the Precision-Recall Curve) to evaluate the performance of the model. AUC measures the model's ability to discriminate between positive and negative samples, while AUPR focuses on the balance between precision and recall. Both metrics have values closer to 1, indicating a better model.

c) *Baselines*: We select a variety of models based on different algorithms for comparative analysis, including:

- Random Forest (RF) [38] was an integrated learning method for classification.
- Support Vector Machines (SVM) [39] was a classic supervised learning technique by directly providing embedding information about drugs and their targets.
- XGBoost [40] fused a decision tree model with a gradient enhancement mechanism to improve both prediction accuracy and computational efficiency.
- GCN [41] was an archetypal semi-supervised learning method. It uses the drug-target association network as input and predicts other potential drug-target pairings.
- Graphical Attention Networks (GAT) [42] was used to capture not only the complex relationships between drugs and targets, but also the deep-level feature representations.
- DTICNN [43] captured the embedded representations of drugs and their targets through heterogeneous networks and infers their interactions with the learned features using denoising autoencoder models.
- GCNMDA [44] explored the correlation between drugs and microorganisms by constructing a network between them.
- MVGCN [45] utilizes the design of multi-view GCN to fully account for the complexity and diversity of biomedical data and improve prediction accuracy.

d) *Hyper meters-Setting*: We use a grid search to identify the optimal hyperparameter settings for each dataset. These settings include the number of deep neural network layers in the Q target and prediction networks, the number of nodes in the input layer (input nodes), the number of nodes in each hidden layer (nodes in hidden layers), and the learning rate (lr), and the value of γ in RL.

Lou's datasets: Deep neural network layers - 4, input nodes - 1500, nodes in hidden layers - 512, 256, 128, 64. lr - 0.001, γ - 0.9.

Yamanishi's datasets: Deep neural network layers - 3, input nodes - 1500, nodes in hidden layers - 512, 256, 128. lr - 0.001, γ - 0.85.

Zheng's datasets: Deep neural network layers - 4, input nodes - 1500, nodes in hidden layers - 512, 256, 128, 64. lr - 0.001, γ - 0.9.

B. Performance Analysis

Tables I, II and III show the results of the prediction accuracy at different scales. It can be concluded.

TABLE I: Results of DTI prediction at different scales on Luo’s dataset.

Models	1:1		1:10	
	AUC	AUPR	AUC	AUPR
RF [38]	0.9016	0.9129	0.9176	0.7156
SVM [39]	0.8586	0.8111	0.8871	0.6078
XGBoost [40]	0.9238	0.9323	0.9311	0.7864
GCN [41]	0.8938	0.8758	0.8617	0.5817
GAT [42]	0.8759	0.8668	0.8525	0.5340
DTICNN [43]	0.9262	0.9340	0.9319	<u>0.7957</u>
GCNMDA [44]	0.9424	0.9347	0.9423	0.6573
MVGCN [45]	0.9042	0.9017	0.9163	0.6959
MuRL-DTI	0.9541	0.9477	0.9513	0.8145

TABLE II: Results of DTI prediction at different scales on Yamanishi’s dataset.

Models	1:1		1:10	
	AUC	AUPR	AUC	AUPR
RF [38]	0.8485	0.8867	0.8475	0.8432
SVM [39]	0.8411	0.8271	0.8071	0.7834
XGBoost [40]	0.9258	0.9213	0.9311	0.7864
GCN [41]	0.8818	0.8271	0.8313	0.6897
GAT [42]	0.8913	0.8578	0.8561	0.7382
DTICNN [43]	0.9012	0.8640	0.8612	0.8057
GCNMDA [44]	0.9214	0.8517	0.8221	0.8461
MVGCN [45]	0.9412	0.8819	0.8983	<u>0.9159</u>
MuRL-DTI	0.9542	0.9317	0.9413	0.9209

- The results on Lou’s dataset show that MuRL-DTI outperforms the baselines. Specifically, compared to the DTICNN model, there is an average improvement of about 2.4% and 5.2% in the AUC and AUPR evaluation metrics, respectively. Compared to the GCNMDA model, the AUC and AUPR metrics are improved by 1.2% and 12.3%, respectively. Compared to the MVGCN model, the performance improvement is even 3.6% and 10.1%, respectively. These results demonstrate the effectiveness of the proposed model. The results from Yamanishi’s and Zheng’s datasets also demonstrate that MuRL-DTI performs best compared to baseline models.
- We compare the performance of MuRL-DTI with the baseline model at different sample ratios. The results from the Lou’s dataset indicate that the AUC and AUPR metrics of MuRL-DTI outperform the baselines at a sample ratio of 1:1, reaching high values of 0.9541 and 0.9477, respectively. The AUC and AUPR of the GCNMDA model are 0.9424 and 0.9347, which are 1.2% and 1.3% lower than MuRL-DTI. Furthermore, the model maintains its superior performance when the sample

TABLE III: Results of DTI prediction at different scales on Zheng’s dataset.

Models	1:1		1:10	
	AUC	AUPR	AUC	AUPR
RF [38]	0.9489	0.9481	0.8935	0.8001
SVM [39]	0.9181	0.8824	0.9033	0.8249
XGBoost [40]	0.9440	0.9319	0.8909	0.8303
GCN [41]	0.9273	0.9100	0.8898	0.6926
GAT [42]	0.9186	0.8933	0.8729	0.7527
DTICNN [43]	0.9187	0.9132	0.8818	<u>0.8817</u>
GCNMDA [44]	0.5405	0.3727	0.4963	0.3422
MVGCN [45]	0.9024	0.8994	0.8495	0.8627
MuRL-DTI	0.9591	0.9612	0.9323	0.9019

ratio is increased to 1:10, further confirming its robustness and efficiency.

C. Ablation Studies

We evaluate the contribution of the proposed multimodal fusion (MF) module and the DQN-RL module to the performance of the model. The results of the ablation experiments are presented in Tables IV, V, and VI. The results show that both proposed modules significantly improve the overall performance of the model, demonstrating their effectiveness.

TABLE IV: Ablation Studies on Lou’s datasets

Models	1:1		1:10	
	AUC	AUPR	AUC	AUPR
DQN-RL	0.9321	0.9294	0.9295	0.7824
DQN-RL + MF	0.9541	0.9477	0.9513	0.8145

TABLE V: Ablation Studies on Yamanishi’s datasets

Models	1:1		1:10	
	AUC	AUPR	AUC	AUPR
DQN-RL	0.9234	0.9091	0.9291	0.9027
DQN-RL + MF	0.9542	0.9317	0.9413	0.9209

TABLE VI: Ablation Studies on Zheng’s datasets

Models	1:1		1:10	
	AUC	AUPR	AUC	AUPR
DQN-RL	0.9134	0.9294	0.9192	0.8747
DQN-RL + MF	0.9591	0.9612	0.9323	0.9019

D. Parameter Sensitivity Analysis

We evaluate the effect of the number of deep neural network layers on model performance. Fig. 4 shows that as the number of layers increases, there is a threshold beyond which the positive impact on performance becomes insignificant, leading to increased complexity in the model.

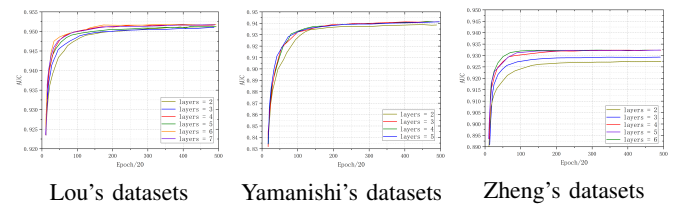


Fig. 4: The effect of the number of deep neural network layers.

IV. CONCLUSION

In this paper, we propose a multimodal fusion RL model called MuRL-DTI for DTI prediction. The model employs a DQN-based RL framework to address data sparsity and cold-start problems by leveraging existing knowledge to facilitate learning in unknown drugs and target interaction prediction. We incorporate two biological features: sequence features and structural features of compounds to enrich the environmental information in reinforcement learning and improve prediction accuracy. We design positive and negative sample distribution ratios of 1:1 and 1:10 on three standard DTI datasets for testing. Experimental results show that our model outperforms baseline models, highlighting the effectiveness of our proposed method. Future work will explore the application of reinforcement learning in other biomedical domains.

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